

Write Up PicoCTF

Forensics

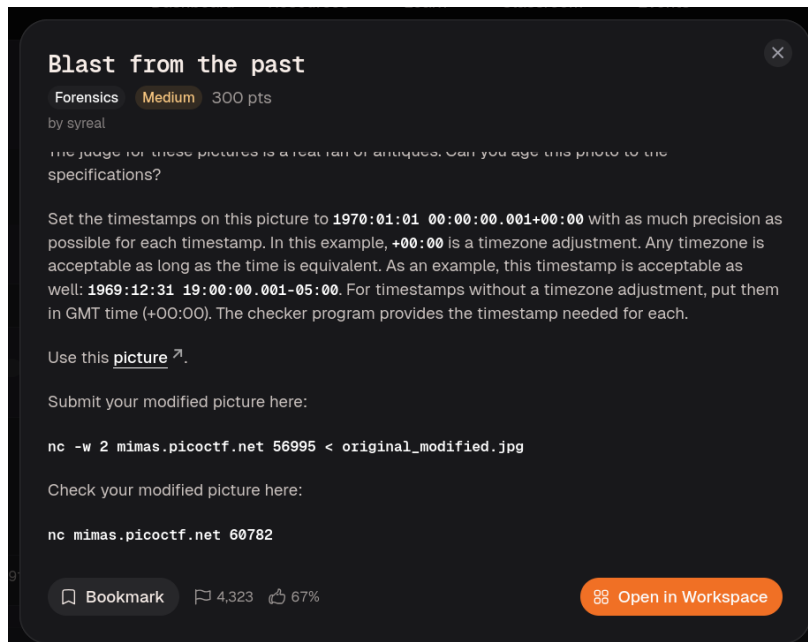


SMK NEGERI 1 PEMALANG

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Daftar Isi

1. Blast From The Past



Blast from the past Forensics Medium 300 pts
by syreal

The judge for these pictures is a real fan of antiques. Can you age this photo to the specifications?

Set the timestamps on this picture to **1970:01:01 00:00:00.001+00:00** with as much precision as possible for each timestamp. In this example, **+00:00** is a timezone adjustment. Any timezone is acceptable as long as the time is equivalent. As an example, this timestamp is acceptable as well: **1969:12:31 19:00:00.001-05:00**. For timestamps without a timezone adjustment, put them in GMT time (+00:00). The checker program provides the timestamp needed for each.

Use this [picture](#).

Submit your modified picture here:

```
nc -w 2 mimas.picoctf.net 56995 < original_modified.jpg
```

Check your modified picture here:

```
nc mimas.picoctf.net 60782
```

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Hint : Exiftool is really good at reading metadata, but you might want to use something else to modify it.

Soal memberikan kita sebuah file “original.jpg” dengan tugas: ubah semua timestamp yang ada di metadata tersebut menjadi 1970:01:01 00:00:00.001+00:00. Tanggal ini disebut Unix Epoch – titik nol perhitungan waktu di sistem computer. Target soal ini adalah 1 milidetik setelah epoch, bukan pas epoch (0), kemungkinan supaya program pengecek otomatis tidak menganggapnya sebagai nilai kosong.

Mari kita cek metadatanya dengan menggunakan command **exiftool**

```

[naren@parrot]~[~/CTF/picoCTF/foren/blastfromthepast]
$ls
original.jpg
[naren@parrot]~[~/CTF/picoCTF/foren/blastfromthepast]
$exiftool original.jpg
ExifTool Version Number      : 13.25
File Name                    : original.jpg
Directory                    : .
File Size                     : 2.9 MB
File Modification Date/Time   : 2024:03:13 13:45:08-04:00
File Access Date/Time        : 2026:06:28 05:56:31-04:00
File Inode Change Date/Time   : 2026:06:28 05:56:15-04:00
File Permissions              : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Little-endian (Intel, II)
Image Description             :
Make                         : samsung
Camera Model Name             : SM-A326U
Orientation                   : Rotate 90 CW
X Resolution                  : 72
Y Resolution                  : 72
Resolution Unit               : inches
Software                      : MediaTek Camera Application
Modify Date                   : 2023:11:20 15:46:23
Y Cb Cr Positioning          : Co-sited
Exposure Time                 : 1/24
F Number                      : 1.8
Exposure Program              : Program AE
ISO                           : 500
Sensitivity Type              : Unknown
Recommended Exposure Index    : 0
Exif Version                  : 0220
Date/Time Original            : 2023:11:20 15:46:23
Create Date                   : 2023:11:20 15:46:23
Components Configuration      : Y, Cb, Cr, -
Shutter Speed Value           : 1/24
Aperture Value                : 1.9
Brightness Value              : 3

```

```

File Edit View Search Terminal Help
Sub Sec Time Original          : 703
Sub Sec Time Digitized        : 703
Flashpix Version               : 0100
Color Space                    : sRGB
Exif Image Width               : 4000
Exif Image Height              : 3000
Interoperability Index        : R98 - DCF basic file (sRGB)
Interoperability Version       : 0100
Exposure Mode                  : Auto
White Balance                  : Auto
Digital Zoom Ratio             : 1
Focal Length In 35mm Format    : 25 mm
Scene Capture Type             : Standard
Compression                    : JPEG (old-style)
Thumbnail Offset               : 1408
Thumbnail Length               : 64000
Image Width                   : 4000
Image Height                   : 3000
Encoding Process               : Baseline DCT, Huffman coding
Bits Per Sample                : 8
Color Components               : 3
Y Cb Cr Sub Sampling           : YCbCr4:2:0 (2 2)
Time Stamp                     : 2023:11:20 15:46:21.420-05:00
MCC Data                       : United States / Guam (310)
Aperture                       : 1.8
Image Size                     : 4000x3000
Megapixels                     : 12.0
Scale Factor To 35 mm Equivalent: 5.4
Shutter Speed                  : 1/24
Create Date                    : 2023:11:20 15:46:23.703
Date/Time Original             : 2023:11:20 15:46:23.703
Modify Date                    : 2023:11:20 15:46:23.703
Thumbnail Image                : (Binary data 64000 bytes, use -b option to extract)
Circle Of Confusion            : 0.006 mm
Field Of View                  : 71.5 deg
Focal Length                   : 4.6 mm (35 mm equivalent: 25.0 mm)
Hyperfocal Distance            : 2.13 m
Light Value                    : 4.0

```

Dari keterangan soal dan exiftool, kita harus mengubah metadata CreateDate, Date/Time Original dan Modify Date menjadi 1970:01:01 00:00:00.001+00:00

Kita bisa mengubah metadata tersebut dengan menggunakan command

```
$exiftool -ModifyDate="1970:01:01 00:00:00.001+00:00" -DateTimeOriginal="1970:01:01 00:00:00.001+00:00" -CreateDate="1970:01:01 00:00:00.001+00:00" original.jpg
```

```
Create Date           : 1970:01:01 00:00:00.703
Date/Time Original    : 1970:01:01 00:00:00.703
Modify Date           : 1970:01:01 00:00:00.703
Thumbnail Image       : (Binary data 64000 bytes, use -b option to extract)
Circle Of Confusion   : 0.006 mm
Field Of View          : 71.5 deg
Focal Length           : 4.6 mm (35 mm equivalent: 25.0 mm)
Hyperfocal Distance   : 2.13 m
```

Bisa kita lihat bahwa metadatnya sudah berubah.

Mari mengirim file tersebut menggunakan command

```
nc -w 2 mimas.picocft.net 56995 < original.jpg
```

Setelah itu mari kita cek menggunakan command

```
nc mimas.picocft.net 60782
```

```

[naren@parrot]~[~/CTF/picoCTF/foren/blastfrom
$nc mimas.picoctf.net 60782
MD5 of your picture: caac017be641a5201f29bd0c004fe0d7 test.out

Checking tag 1/7
Looking at IFD0: ModifyDate
Looking for '1970:01:01 00:00:00'
Found: 1970:01:01 00:00:00
Great job, you got that one!

Checking tag 2/7
Looking at ExifIFD: DateTimeOriginal
Looking for '1970:01:01 00:00:00'
Found: 1970:01:01 00:00:00
Great job, you got that one!

Checking tag 3/7
Looking at ExifIFD: CreateDate
Looking for '1970:01:01 00:00:00'
Found: 1970:01:01 00:00:00
Great job, you got that one!

Checking tag 4/7
Looking at Composite: SubSecCreateDate
Looking for '1970:01:01 00:00:00.001'
Found: 1970:01:01 00:00:00.703
Oops! That tag isn't right. Please try again.

```

Sepertinya kita belum mengubah SubSecCreateDate, SubSecDateTimeOriginal dan SubSecModifyDate, kita bisa mengubahnya dengan menggunakan command :

```

$exiftool -SubSecCreateDate="1970:01:01 00:00:00.001+00:00" -
SubSecModifyDate="1970:01:01 00:00:00.001+00:00" -
SubSecDateTimeOriginal="1970:01:01 00:00:00.001+00:00" -
SubSecTimeDigitized="001" original.jpg

```

Gas kita kirim lagi

Sepertinya metadatanya tertulis sebagai teks ASCII berisi angka (bukan integer biner), yang jika kita konversikan akan menjadi 20 November 2023

Kita cari byte 1700513181420 dengan menggunakan command ghex.

\$ghex original.jpg

```
31 37 30 30 35 31 33 31 38 31 34 32 30 1700513181420

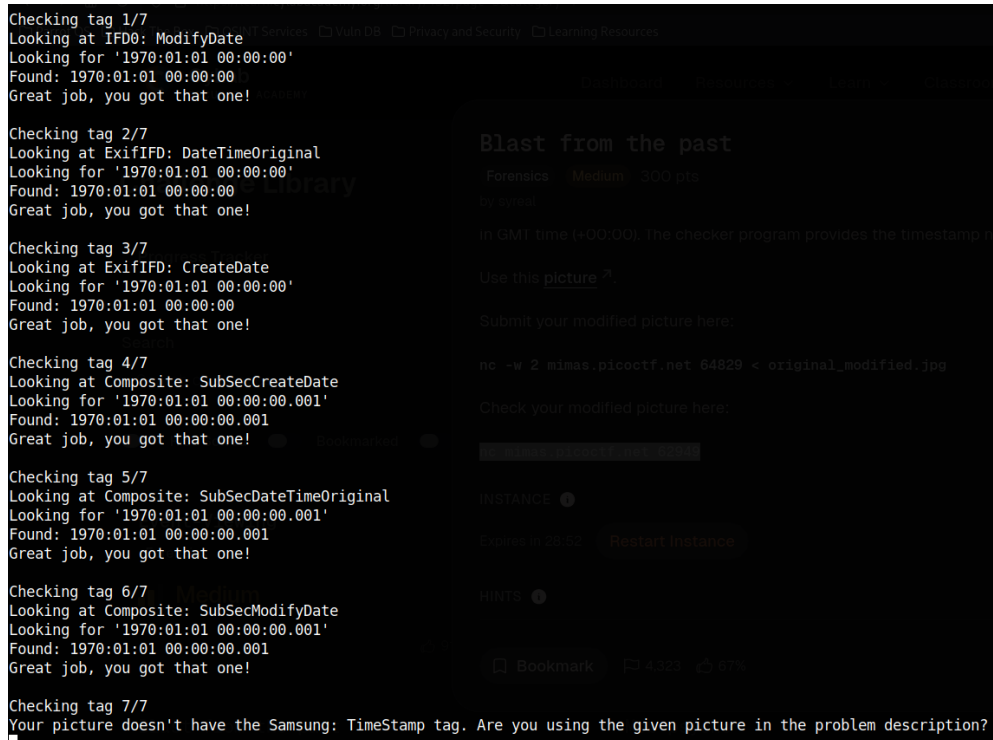
Find Next Find Previous Clear

002B81E0 A7 71 FC 5F EF 3F FE 80 B5 20 54 D8 0C 68 CE 24 .q._.?... T..h.$
002B81F0 44 65 C0 91 9B A1 3E 9F 4E B5 55 64 8B 78 2C B2 De....>.N.Ud.x,.
002B8200 46 C8 49 2F 8E E3 9C 92 7A 8A D0 BB FF 00 8F 18 F.I/....z.....
002B8210 3E AB FC AB 2E 5F F5 72 FF 00 D7 03 FC 85 20 1C >...._r.....
002B8220 4F 98 03 19 94 A8 73 96 53 C6 5B BF D6 A3 78 C3 O.....s.S.[...x.
002B8230 A3 64 ED 7C 06 24 0C 1E 07 5A AF 6B FF 00 1E A7 .d.|.$...Z.k....
002B8240 FE BA AD 5D 6F BC FF 00 F5 EE 68 02 A3 2B ED 02 ...]o.....h..+..
002B8250 46 5D C5 4F EF 48 E0 9F FE BD 56 57 0B 21 77 79 F].O.H....VW.!wy
002B8260 01 9A 3E 10 B6 40 3D 39 5F 5E A6 AF 5D 7F C7 B4 ..>..@=9_^..]...
002B8270 5F F0 1A C7 B8 FF 00 5E BF F5 F0 3F 91 A0 07 4F _.....^...?...0
002B8280 2B B5 B3 36 F5 52 F2 06 EA 71 8F E9 EB F5 AA BE +..6.R...q.....
002B8290 73 94 8D DE 50 EE E0 B4 D8 3E 87 F9 9A 74 BF F1 s...P....>...t..
002B82A0 E2 FF 00 EE AF F3 AA 0B FE AD BE 8B 4D 6E 60 6C .....Mn`l
002B82B0 BB F3 1A C9 E4 A1 8F E6 F3 CF 7C 8E 83 EA 28 77 .....|... (w
002B82C0 3B 11 72 D9 90 EF 04 9C 82 3D 47 B6 47 4A A9 7B ;.r.....=G.GJ.{
002B82D0 F7 63 FA 2F FE 82 2A CB FF 00 AB D3 BF EB C5 BF .c./...*.....
002B82E0 F4 33 5D 50 DC C6 6D DC FF D9 00 00 01 0A 0E 00 .3]P..m.....
002B82F0 00 00 49 6D 61 67 65 5F 55 54 43 5F 44 61 74 61 ..Image UTC Data
002B8300 31 37 30 30 35 31 33 31 38 31 34 32 30 00 00 A1 1700513181420...
002B8310 0A 08 00 00 00 4D 43 43 5F 44 61 74 61 33 31 30 ....MCC Data310
002B8320 00 00 61 0C 18 00 00 00 43 61 6D 65 72 61 5F 43 a...Camera C

Offset: 0x2B8300
```

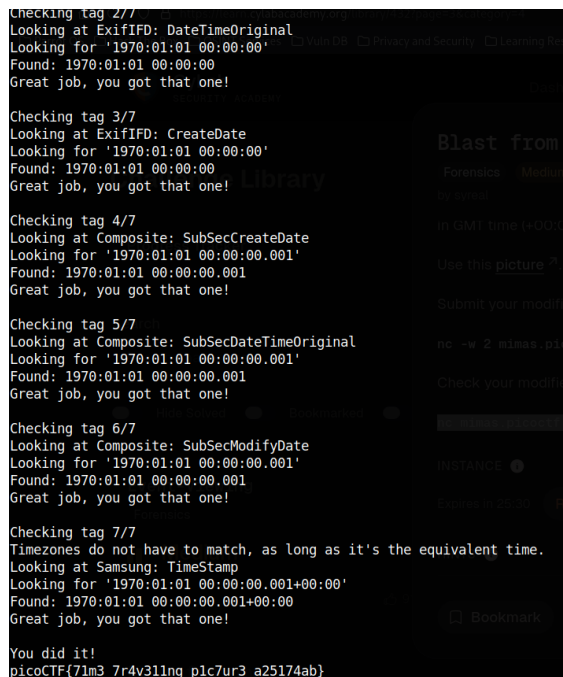
Kita harus mengubah byte 1700513181420 menjadi nol semua.

Setelah diubah, kita simpan filenya dan gas kirim lagi.



Hmmzz sepertinya kita harus mengubah byte 0000000000000000 menjadi 0000000000000001

Gas kita ubah menggunakan command ghex yang tadi.



VOILA FLAG DITEMUKANNNNN

picoCTF{71m3_7r4v311ng_p1c7ur3_a25174ab}